

# ECMA 31370, CUPED White Paper

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## Executive Summary

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## White Paper

### [1] State the Problem and Ask

Hulu is a subscription-based video streaming service that runs A/B tests to evaluate product changes such as recommendation algorithms, UI redesigns, and content curation strategies. All experiments are currently analyzed using difference-in-means (DM). The problem is that most product changes produce small lifts (1 to 3% relative to baseline), and user-level streaming behavior is highly variable, with weekly hours ranging from zero to 30+. The noise in the data regularly overwhelms the treatment signal, so experiments that produce real improvements often end up with confidence intervals that include zero. Kohavi and Thomke (2017) document that this is a common pattern across the industry: variance, not sample size or idea quality, is the binding constraint in most online experiments.

We ask whether adopting CUPED (Controlled-experiment Using Pre-Experiment Data) can solve this problem. Our proof-of-concept simulation (Section 7) shows that it can: with Hulu's data characteristics, the naive DM estimator fails to detect a true 0.10 hrs/week treatment effect, while CUPED with a 2-week or longer pre-experiment window detects it.

### [2] Make a Recommendation

We recommend that Hulu adopt CUPED as the default analysis method for all A/B tests. The covariate should be pre-experiment weekly streaming hours (which Hulu already logs), with a default window of 4 weeks. For new users without history (about 20% of the experiment population), we follow Booking.com's COALESCE approach and leave their metrics unadjusted (Jackson, 2018). CUPED should be built into the analysis pipeline as a default rather than an opt-in tool, as Meta has done across hundreds of thousands of concurrent experiments (Meakin & Sangwan, 2024).

Our simulation (Section 7) and industry benchmarks suggest 40 to 60% variance reduction for existing users. Booking.com reported reaching significance roughly twice as fast after adopting CUPED (Jackson, 2018). For Hulu, this translates to either detecting smaller effects at the same sample size, or reaching conclusions in half the time.

### [3] Describe the CUPED Methodology

CUPED (Controlled Experiments Using Pre-Experiment Data) is a variance-reduction methodology used in randomized experiments and A/B testing. CUPED utilizes data collected prior to experimentation to reduce variability in treatment effect estimates, allowing researchers to detect smaller effects or achieve similar statistical power with smaller sample sizes. Because of this, CUPED has proven to be an effective tool for causal analysis in industry since its inception at Microsoft,.

CUPED seeks to reduce variability by introducing a control random variable that is both unaffected by the treatment and correlated with the experimental outcome variable. A commonly used metric that accomplishes this is pre-experimental data. For example, in an experiment where a team is seeking to understand the effect a new sports campaign has on equipment sales, prior spending data can be used as the control variable. Because  $X_i$  is measured before treatment assignment, treatment cannot affect it, ensuring the adjustment does not introduce confounding bias. Additionally, pre-experimental data is generally correlated with future spending trends, which are often persistent behaviors for individuals.

Formally, let  $Y_i$  be the experimental outcome and  $X_i$  be the pre-experiment metric. Then, CUPED constructs the adjusted estimator:

$$Y_i^{CUPED} = Y_i - \theta X_i + \theta E[X_i] = Y_i - \theta(X_i - E[X_i])$$

Then, we find the optimal  $\theta$  such that:

$$\theta^* = \frac{Cov(Y, X)}{Var(X)}$$

This removes the component of the outcome's variability that can be predicted using the pre-treatment covariate. When substituting this into the variance for  $Y_i^{CUPED}$ , we get:

$$Var(Y^{CUPED}) = (1 - \rho^2)Var(Y)$$

Hence, the more correlated  $X_i$  and  $Y_i$  are, the greater the variance reduction achieved by CUPED. CUPED is especially useful in large-scale experimentation settings where even small improvements in statistical precision can translate into substantial savings in experimentation costs or meaningful increases in revenue.

## [4] Compare CUPED to Other Methodologies

## [5] Report on CUPED Adoption Elsewhere

Netflix was among the first to evaluate CUPED in production. Xie and Aurisset (2016) compared it against stratified sampling and post-stratification using previous-week streaming hours as the covariate. For existing users with viewing history, all three methods achieved roughly 40% variance reduction at  $\rho = 0.7$  to 0.8. Stratified sampling, however, degraded from about 45% to 15% in Netflix’s distributed multi-machine environment because balanced assignment could not be maintained across servers; CUPED, applied after data collection, was unaffected. Netflix concluded that post-assignment methods are preferable and that the choice of covariate matters more than the choice of method. They also found that new users without viewing history offered little to adjust on:  $\rho$  dropped to 0.2–0.4 and variance reduction was negligible.

Booking.com (Jackson, 2018) tackled the missing-data problem that Netflix flagged. Booking.com processes over 1.5 million room-night reservations per day, but with a conversion rate of about 2%, detecting even a 1% relative improvement requires over 15 million users under standard methods, so variance reduction is especially valuable at their scale. Rather than excluding first-time visitors who lack pre-experiment data, they implemented a one-line SQL COALESCE pattern that leaves those users’ metrics unadjusted. Mathematically, this is equivalent to imputing missing covariates with the sample mean, which zeroes out the adjustment term. In a live experiment on a new calendar interface, the CUPED-adjusted metric reached significance around day 5–6, while the unadjusted metric did not reach significance until day 11–12, roughly halving the time needed to call the experiment. The COALESCE approach has since become standard practice at other companies as well.

Faire (Ritchie & Wong, 2021), a B2B wholesale marketplace with sample sizes in the tens of thousands, ran the most systematic study of pre-experiment window length. They tested windows of 1, 2, 4, 8, and 16 weeks: variance reduction rose from about 22% at 1 week to 45% at 8 weeks and plateaued at 50% by 16 weeks. Faire settled on 8 weeks as their default, noting that 90% of consecutive orders on their platform occur within 28 days. They also combined CUPED with outlier capping at 5 standard deviations and found the two techniques complementary, since outlier capping limits the influence of extreme values while CUPED removes baseline user-level variation.

DoorDash (Zhang & Tang, 2020) illustrates a different dimension of CUPED’s value. Before adopting variance reduction, the platform could run only about 2 concurrent experiments on its consumer app because high variance in order-value metrics required large sample sizes that consumed most of the available traffic. After implementing CUPED alongside experiment parallelization, they achieved a 4X increase in experiment capacity. DoorDash also found that the type of covariate matters: user-level behavioral data (such as order history) reduced the required sample size by 10–20%, while categorical attributes like platform (iOS vs Android) and market (city) contributed only 2–4%. The lesson is that investing in user-level pre-experiment features pays off far more than relying on broad demographic segments.

CUPED has since scaled well beyond these early adopters. Meta applies it across hundreds of thousands of concurrent experiments under the internal name “Mean2.0,” with over 30% average confidence interval reduction (Meakin & Sangwan, 2024). Commercial platforms including Statsig and Eppo now ship CUPED as an enabled-by-default feature (Sexauer, 2024).

These experiences inform several design choices for Hulu. Netflix used weekly streaming hours as its core metric and observed

$\rho \approx 0.7$  with  $\sim 40\%$  variance reduction, which provides a direct benchmark for what Hulu can expect. The new-user limitation Netflix documented applies as well, given that roughly 20% of Hulu’s experiment participants are new subscribers without viewing history. Booking.com’s COALESCE approach offers a ready-made solution for including these users without discarding them. Faire’s window-length analysis suggests that a 4-week default is a natural starting point for a platform where viewing habits follow a weekly rhythm, and that extending beyond 4 weeks yields diminishing returns. DoorDash’s finding on covariate quality reinforces the case for using streaming hours (a user-level behavioral metric) rather than demographic segments as the primary covariate. Our simulation in Section 7 tests these choices under Hulu-like parameters and produces results consistent with the patterns reported above.

## [6] Describe Implementation

## [7] Report on a POC

We simulate an A/B test on Hulu to illustrate how CUPED reduces estimator variance. The platform is testing a new recommendation algorithm, and the primary metric is weekly streaming hours per user. Full simulation code is provided in Appendix A.

### Data Generating Process

Each user  $i$  has a stable baseline streaming rate  $\mu_i \sim \text{LogNormal}(\log 7, 0.7)$ , producing a right-skewed distribution with median around 7 hours per week. Weekly streaming hours are generated as:

$$\text{hours}_{it} = \max(0, \mu_i + \tau \cdot W_i + \varepsilon_{it})$$

where  $W_i \in \{0, 1\}$  is the treatment indicator,  $\tau = 0.10$  hours/week is the true treatment effect, and  $\varepsilon_{it} \sim N(0, 5^2)$  is independent week-to-week noise. The floor at zero ensures non-negative hours. Light users with small  $\mu_i$  naturally produce zero-hours weeks when noise pushes them below zero, and because the same users tend to be light across all weeks, these zeros are correlated across pre and post periods, preserving realistic pre-post correlation without any additional mechanism.

We generate 50,000 users (80% existing, 20% new), 4 weeks of pre-experiment data, and 1 week of experiment data. New users have no pre-experiment observations.

### Pre-Post Correlation and Method Comparison

The pre-post correlation for existing users increases from 0.697 (1-week window) to 0.756 (2 weeks) and 0.794 (4 weeks), because averaging over more weeks smooths out noise and yields a more stable estimate of each user’s baseline. The improvement from 1 to 2 weeks is larger than from 2 to 4 weeks, consistent with diminishing returns reported by Faire (Ritchie & Wong, 2021).

Since CUPED’s theoretical variance reduction equals  $\rho^2$ , these correlations imply that extending the covariate window from 1 to 4 weeks raises variance reduction from roughly 49% to 63%.

We compare four estimators: naive difference-in-means, first differences ( $Y_{post} - X_{pre}$ , equivalent to CUPED with  $\theta$  forced to 1), CUPED (data-driven optimal  $\hat{\theta}$ ), and OLS regression adjustment. For new users with missing pre-experiment data, we apply the Booking.com COALESCE approach and leave their outcome unadjusted (Jackson, 2018). Full numerical results are reported in Appendix B (Tables B1 and B2).

## Results

CUPED and OLS regression adjustment produce identical point estimates, standard errors, and confidence intervals across all window lengths. This follows from the fact that the two methods are mathematically equivalent when a single covariate is used: CUPED’s  $\hat{\theta} = \text{Cov}(Y, X)/\text{Var}(X)$  is the OLS slope on  $X$ , so the adjusted outcomes are the same.

First differences performs slightly worse than CUPED because it fixes  $\theta = 1$ , while the data-driven  $\hat{\theta}$  ranges from about 0.69 (1-week) to 0.90 (4-week). The gap is largest at short windows: 39.1% variance reduction for first differences versus 48.5% for CUPED at 1 week. As the window lengthens and  $\hat{\theta}$  approaches 1, the two methods converge (62.4% versus 63.1% at 4 weeks).

Naive DM fails to detect the 0.10 hrs/week treatment effect at any window length because its confidence interval always includes zero. CUPED and first differences both reach significance at the 2-week and 4-week windows. When new users (20% of the sample) are included via the COALESCE approach, CUPED with a 4-week window still achieves 50.2% variance reduction at the population level, compared to 63.1% for existing users alone. New users in isolation have a standard error of 0.1666, roughly twice that of the full sample (0.0736), confirming the Netflix finding that CUPED provides no benefit for users without historical data (Xie & Aurisset, 2016).

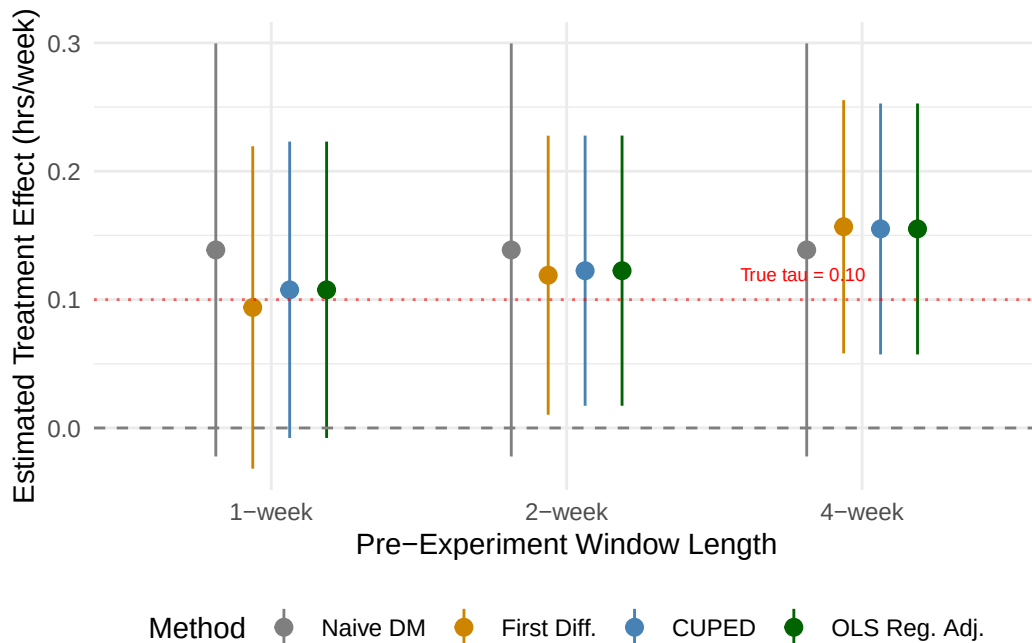


Figure 1: 95% confidence intervals by method and window length (existing users). Dashed gray line: zero. Dotted red line: true

effect. CUPED and OLS overlap exactly.

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## Appendix

### [A] Simulation Code

```
library(tidyverse)

set.seed(42)

# -- Parameters --
N <- 50000
new_user_frac <- 0.20
log_mean <- log(7)
log_sd <- 0.7
noise_sd <- 5.0
tau <- 0.10
n_pre_weeks <- 4

# -- Generate Users --
n_new <- round(N * new_user_frac)
n_existing <- N - n_new

users <- tibble(
  user_id = 1:N,
  is_new = c(rep(TRUE, n_new), rep(FALSE, n_existing)),
  baseline_mu = rlnorm(N, meanlog = log_mean, sdlog = log_sd),
  treatment = sample(c(0L, 1L), N, replace = TRUE)
)

# -- Generate Weekly Streaming Hours --
generate_week <- function(baseline_mu, treatment_indicator = NULL, tau_val = 0) {
  n <- length(baseline_mu)
  noise <- rnorm(n, mean = 0, sd = noise_sd)
  hours <- baseline_mu + noise
```

```

if (!is.null(treatment_indicator)) {
  hours <- hours + tau_val * treatment_indicator
}
return(pmax(hours, 0))
}

pre_weeks <- map(1:n_pre_weeks, ~ generate_week(users$baseline_mu))
names(pre_weeks) <- paste0("pre_week_", 1:n_pre_weeks)
post_week <- generate_week(users$baseline_mu, users$treatment, tau)

# -- Assemble Dataset --
sim_data <- users %>%
  bind_cols(as_tibble(pre_weeks)) %>%
  mutate(
    post_week = post_week,
    pre_1wk = pre_week_4,
    pre_2wk = (pre_week_3 + pre_week_4) / 2,
    pre_4wk = (pre_week_1 + pre_week_2 + pre_week_3 + pre_week_4) / 4
  ) %>%
  mutate(
    pre_1wk = if_else(is_new, NA_real_, pre_1wk),
    pre_2wk = if_else(is_new, NA_real_, pre_2wk),
    pre_4wk = if_else(is_new, NA_real_, pre_4wk)
  )

# -- CUPED Adjustment (Booking.com COALESCE for missing data) --
cuped_adjust <- function(y, x) {
  has_x <- !is.na(x)
  theta <- cov(y[has_x], x[has_x]) / var(x[has_x])
  x_mean <- mean(x[has_x])
  y_adj <- y
  y_adj[has_x] <- y[has_x] - theta * (x[has_x] - x_mean)
  return(list(y_adj = y_adj, theta = theta))
}

# -- Four-Method Comparison --

```

```

run_test_all <- function(data, pre_col, label) {
  y <- data$post_week
  x <- data[[pre_col]]
  w <- data$treatment
  has_x <- !is.na(x)

  y_t <- y[w == 1]; y_c <- y[w == 0]
  n_t <- length(y_t); n_c <- length(y_c)

  # Naive DM
  naive_est <- mean(y_t) - mean(y_c)
  naive_se <- sqrt(var(y_t) / n_t + var(y_c) / n_c)

  # First Differences (theta = 1)
  fd <- y; fd[has_x] <- y[has_x] - x[has_x]
  fd_t <- fd[w == 1]; fd_c <- fd[w == 0]
  fd_est <- mean(fd_t) - mean(fd_c)
  fd_se <- sqrt(var(fd_t) / n_t + var(fd_c) / n_c)

  # CUPED (optimal theta)
  res <- cuped_adjust(y, x)
  ya <- res$y_adj
  ya_t <- ya[w == 1]; ya_c <- ya[w == 0]
  cuped_est <- mean(ya_t) - mean(ya_c)
  cuped_se <- sqrt(var(ya_t) / n_t + var(ya_c) / n_c)

  # OLS Regression Adjustment
  df_ols <- tibble(y = y, w = w, x = x) %>% filter(!is.na(x))
  mod <- lm(y ~ w + x, data = df_ols)
  ols_est <- coef(mod)["w"]
  ols_se <- summary(mod)$coefficients["w", "Std. Error"]

  cat(sprintf("[%s]\n", label))
  cat(sprintf("  Naive DM:      est=%.4f SE=%.4f VR= 0.0%%\n", naive_est, naive_se))
  cat(sprintf("  First Diff:   est=%.4f SE=%.4f VR=%5.1f%%\n",
              fd_est, fd_se, (1 - fd_se^2/naive_se^2)*100))

```



```

cat(sprintf("  CUPED:          est=%.4f SE=%.4f VR=%5.1f%%\n",
           cuped_est, cuped_se, (1 - cuped_se^2/naive_se^2)*100))

cat(sprintf("  OLS Reg Adj:  est=%.4f SE=%.4f VR=%5.1f%%\n",
           ols_est, ols_se, (1 - ols_se^2/naive_se^2)*100))
}

```

```

existing <- sim_data %>% filter(!is_new)
run_test_all(existing, "pre_1wk", "Existing, 1-week")
run_test_all(existing, "pre_2wk", "Existing, 2-week")
run_test_all(existing, "pre_4wk", "Existing, 4-week")
run_test_all(sim_data, "pre_1wk", "All users, 1-week")
run_test_all(sim_data, "pre_4wk", "All users, 4-week")

```

## [B] Detailed Simulation Results

Table 1: Table B0: Pre-post correlation and theoretical variance reduction by window length (existing users).

| Pre-Experiment Window | Correlation ( $\rho$ ) | Theoretical VR |
|-----------------------|------------------------|----------------|
| 1 week                | 0.697                  | 48.5%          |
| 2 weeks               | 0.756                  | 57.2%          |
| 4 weeks               | 0.794                  | 63.1%          |

Table 2: Table B1: Standard errors and variance reduction by method (existing users, true  $\tau = 0.10$ ).

| Window | Naive SE | FD SE  | CUPED SE | OLS SE | FD VR | CUPED VR | OLS VR |
|--------|----------|--------|----------|--------|-------|----------|--------|
| 1-week | 0.0821   | 0.0641 | 0.0589   | 0.0589 | 39.1% | 48.5%    | 48.5%  |
| 2-week | 0.0821   | 0.0555 | 0.0537   | 0.0537 | 54.4% | 57.2%    | 57.2%  |
| 4-week | 0.0821   | 0.0503 | 0.0499   | 0.0499 | 62.4% | 63.1%    | 63.1%  |

Table 3: Table B2: Estimates and 95% CIs (existing users). \* = significant at 5%. OLS identical to CUPED, omitted.

| Window | Naive DM                | First Diff.              | CUPED                    |
|--------|-------------------------|--------------------------|--------------------------|
| 1-week | 0.139 [-0.0222, 0.2996] | 0.094 [-0.0317, 0.2195]  | 0.108 [-0.0078, 0.2231]  |
| 2-week | 0.139 [-0.0222, 0.2996] | 0.119 [0.0103, 0.2277] * | 0.122 [0.0173, 0.2278] * |
| 4-week | 0.139 [-0.0222, 0.2996] | 0.157 [0.0581, 0.2554] * | 0.155 [0.0573, 0.2528] * |